

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

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Forename(s)

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Candidate signature

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INTERNATIONAL AS CHEMISTRY (9620)

Unit 1: Inorganic 1 and Physical 1

Tuesday 14 May 2019

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- All working must be shown.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

Titanium is an element that has several isotopes.

0 1 . 1

State why titanium is in the d block of the Periodic Table.

[1 mark]

0 1 . 2

The mass number of an isotope of titanium is 49

Deduce the number of **each** of the fundamental particles in this isotope of titanium.

[2 marks]

The abundance of each isotope in a sample of titanium can be determined in a time of flight (TOF) mass spectrometer. The first stage of the process is ionisation by electron impact.

0 1 . 3

Write an equation, including state symbols, to show this ionisation.

[1 mark]

0 1 . 4

Give **two** reasons why the sample must be ionised.

[2 marks]

Reason 1 _____

Reason 2 _____



0 1 . 5

Complete the full electron configuration of a Ti^+ ion.

[1 mark]

 $1s^2$ _____

0 1 . 6

A sample of titanium contains three isotopes, ^{46}Ti , ^{48}Ti and ^{49}Ti , in a ratio of 2:9:1Calculate the relative atomic mass of this sample of titanium.
Give your answer to one decimal place.

[2 marks]

Relative atomic mass _____

Question 1 continues on the next page

Turn over ►



01.7

Another sample of titanium, containing a different isotope, is analysed in the TOF mass spectrometer.

The ion of this isotope takes 2.863×10^{-6} s to travel down the 1.570 m flight tube. The kinetic energy of this titanium ion is 1.248×10^{-14} J

Calculate the mass, in g, of this titanium isotope.

$$t = d \sqrt{\frac{m}{2E}}$$

t = time / s

d = distance / m

m = mass / kg

E = kinetic energy / J

[3 marks]

Mass _____ g

01.8

Use your answer to Question **01.7** to calculate the mass number of the titanium isotope.

Show your working.

The Avogadro constant, $L = 6.022 \times 10^{23} \text{ mol}^{-1}$

(If you were unable to answer Question **01.7** you should use the value 1.010×10^{-22} g
This is not the correct answer.)

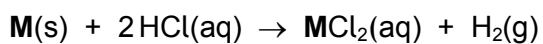
[2 marks]

Mass number _____



0	2
---	---

A metal, **M**, reacts with hydrochloric acid as shown in the equation.



When 0.500 g of **M** reacts with an excess of hydrochloric acid, 314 cm³ of hydrogen are produced, measured at 303 K and 100 kPa

Calculate the relative atomic mass of **M**.
Identify **M**.

The gas constant, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

[5 marks]

Relative atomic mass _____

Identity of **M** _____

5

Turn over ►



0	3
---	---

Metal hydrides are compounds formed when metals react with hydrogen.

0	3	.	1
---	---	---	---

Lithium hydride is an ionic compound. The formula of the hydride ion is H^-

Write an equation for the formation of lithium hydride from its elements.

[1 mark]

0	3	.	2
---	---	---	---

Lithium hydride has a similar structure to sodium chloride.

Draw a diagram to show the three-dimensional (3D) structure of lithium hydride.
You should show at least eight particles.

In your diagram, show the charge on each ion.

[2 marks]



0 3 . 3

At low temperature aluminium hydride exists as molecules of AlH_3

Name the shape of the aluminium hydride molecule and state its bond angle.
Identify the strongest type of force between molecules of AlH_3

[3 marks]

Name of shape _____

Bond angle _____

Strongest force _____

Lithium hydride reacts with aluminium hydride to form the ionic compound
lithium tetrahydridoaluminate(III), LiAlH_4

0 3 . 4

In this reaction, an aluminium hydride molecule reacts with a hydride ion to form the AlH_4^- ion.

Explain how the H^- ion and AlH_3 molecule form a bond to make the AlH_4^- ion.
Name this type of bond.

[2 marks]

Explanation _____

Type of bond _____

0 3 . 5

Lithium tetrahydridoaluminate(III) reacts with oxygen to form lithium oxide, aluminium oxide and water.

Write an equation for this reaction.

[1 mark]



0 4

Sodium hydrogencarbonate (NaHCO_3) reacts with hydrochloric acid as shown.



A student prepared 250 cm^3 of aqueous sodium hydrogencarbonate of concentration $0.150 \text{ mol dm}^{-3}$

0 4 . 1

Calculate the mass, in g, of sodium hydrogencarbonate needed to prepare 250 cm^3 of a $0.150 \text{ mol dm}^{-3}$ solution.

[2 marks]

Mass _____ g

Four 25.0 cm^3 samples of the sodium hydrogencarbonate solution were titrated with hydrochloric acid.

The results are shown in **Table 1**.

Table 1

	Rough	1	2	3
Final burette reading / cm^3	32.0	31.45	31.75	31.50
Initial burette reading / cm^3	0.0	0.40	0.55	0.20
Titre / cm^3				

0 4 . 2

Complete **Table 1**.

Calculate the mean titre.

[2 marks]

Mean titre _____ cm^3



0 4 . 3

Use your answer to Question **04.2** to calculate the concentration, in mol dm^{-3} , of the hydrochloric acid.

[2 marks]

Concentration _____ mol dm^{-3}

0 4 . 4

During each titration, the student rinsed the inside of the conical flask with distilled water.

Explain why this rinsing improves the accuracy of each titre.

[1 mark]

0 4 . 5

In a separate experiment, the student reacted a 75.0 cm^3 sample of $0.200 \text{ mol dm}^{-3}$ sodium hydrogencarbonate solution with hydrochloric acid until the solution was neutral.

The solution formed was evaporated to leave solid sodium chloride.

The yield of sodium chloride was 93%

Calculate the mass, in g, of sodium chloride obtained.

[3 marks]

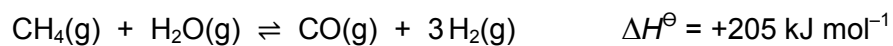
Mass _____ g

10

Turn over ►

0 5

Hydrogen is produced from methane as shown.

**0 5 . 1**

State how the rate of the forward reaction compares with the rate of the reverse reaction at equilibrium.

[1 mark]

0 5 . 2

Explain why an increase in temperature causes an increase in the equilibrium yield of hydrogen at constant pressure.

[2 marks]

0 5 . 3

State and explain the effect of increasing the total pressure on the equilibrium yield of hydrogen at constant temperature.

[3 marks]

Effect on yield _____

Explanation _____



0 5 . 4 Write an expression for the equilibrium constant, K_c , for this equilibrium.

Give its units.

[2 marks]

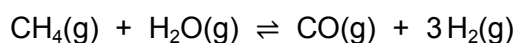
K_c

Units _____

0 5 . 5 A 1.5 mol sample of methane was heated with a 1.0 mol sample of steam in a sealed container.

At equilibrium, the mixture contained 1.2 mol of hydrogen.

Calculate the amount, in moles, of methane and of steam in this equilibrium mixture.



[2 marks]

Amount of methane _____ mol

Amount of steam _____ mol

10

Turn over ►



0	6
---	---

This question is about some test-tube reactions.

0	6	.	1
---	---	---	---

A student adds a few drops of barium chloride solution to some potassium sulfate solution. A white precipitate forms.

Give the formula of the white precipitate.

Write an equation for the formation of the white precipitate.

[2 marks]

Formula of precipitate _____

Equation

0	6	.	2
---	---	---	---

A student adds a few drops of concentrated sulfuric acid to a solid sodium halide. Hydrogen sulfide gas is produced.

Identify the sodium halide.

Write a half-equation for the formation of hydrogen sulfide from sulfuric acid.

State the role of sulfuric acid in the reaction.

[3 marks]

Sodium halide _____

Half-equation

Role of sulfuric acid _____

0	6	.	3
---	---	---	---

A student adds a few drops of silver nitrate solution to a solution of a **different** sodium halide. A precipitate forms that does not dissolve in dilute aqueous ammonia.

Identify the sodium halide.

Write an ionic equation, including state symbols, for the formation of the precipitate.

[2 marks]

Sodium halide _____

Ionic equation



0 6 . 4

A student adds a few drops of silver nitrate solution to another sodium halide solution.
No visible change is seen.

Give the formula of this sodium halide.

[1 mark]

8

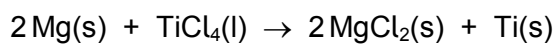
Turn over for the next question

Turn over ►



07

Magnesium is used in the extraction of titanium from titanium chloride as shown.



07.1

Suggest why magnesium is used in this reaction rather than a cheaper metal such as iron.

[1 mark]

Some standard enthalpies of formation are given in **Table 2**.

Table 2

Substance	$\Delta_f H^\ominus / \text{kJ mol}^{-1}$
$\text{TiCl}_4\text{(l)}$	-750
$\text{MgCl}_2\text{(s)}$	-642

07.2

State why the enthalpies of formation of magnesium and titanium do not need to be included in the table.

[1 mark]

07.3

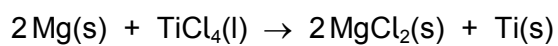
Give the meaning of the term enthalpy change.

[1 mark]



0 7 . 4

Use the data in **Table 2** to calculate the enthalpy change, in kJ mol^{-1} , for the reaction of magnesium with titanium chloride.

**[2 marks]**Enthalpy change _____ kJ mol^{-1}

5

Turn over for the next question**Turn over ►**

0 8

This question is about the elements in Period 3 of the Periodic Table.

0 8 . 1

Identify the Period 3 element from Na to Cl that has the largest atomic radius.

Explain your answer.

[3 marks]

Element _____

Explanation _____

0 8 . 2

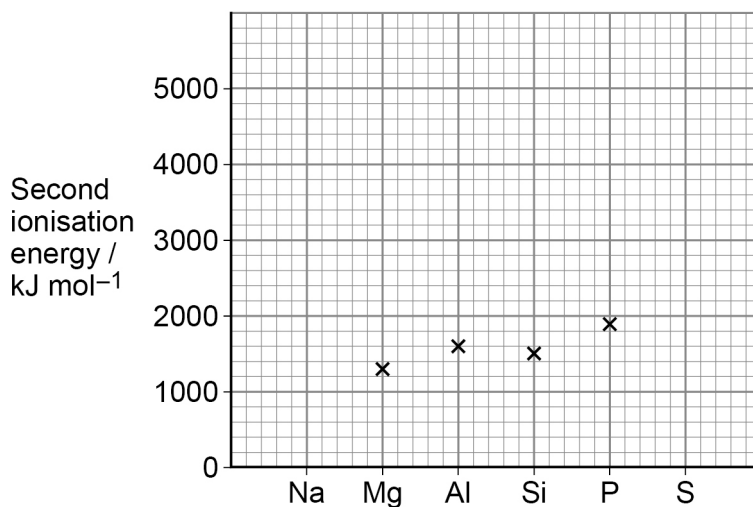
Write an equation, including state symbols, to represent the process that occurs when the **second** ionisation energy of magnesium is measured.

[1 mark]



Figure 1 shows the **second** ionisation energies of some Period 3 elements.

Figure 1



0 8 . 3 Draw a cross on **Figure 1** to show the **second** ionisation energy of sodium.

[1 mark]

0 8 . 4 Draw a cross on **Figure 1** to show the **second** ionisation energy of sulfur.

[1 mark]

Question 8 continues on the next page

Turn over ►



0 8 . 5

Explain why the **second** ionisation energy of silicon is lower than the **second** ionisation energy of aluminium.

[2 marks]

0 8 . 6

An ion of silicon has the electron configuration $1s^2 2s^2 2p^6 3s^1$

Which ionisation of silicon produces this ion?

[1 mark]

Tick (✓) **one** box.

First ionisation

☐

Second ionisation

☐

Third ionisation

☐

9**END OF QUESTIONS**

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Question number	Additional page, if required. Write the question numbers in the left-hand margin.



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2 4



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